

2 Air pollution due to motor vehicles is an environmental issue that needs constant monitoring.

Lichens consist of protocists and fungi living in close association. The protocists and fungi each benefit from this association. Lichens are found in a large range of habitats, including on the surface of tree trunks (tree bark), as shown in Fig. 2.1.



Fig. 2.1

Lichens are sensitive to air pollution. The abundance of lichens decreases with increasing levels of air pollution and so lichens can be used to monitor levels of air pollution.

Some scientists investigated the relationship between distance from a road and levels of air pollution. The scientists investigated this relationship by measuring the abundance of lichens on the tree trunks in a woodland beside a busy road.

The scientists used the grid shown in Fig. 2.2 to determine the abundance of lichens on the tree trunks.

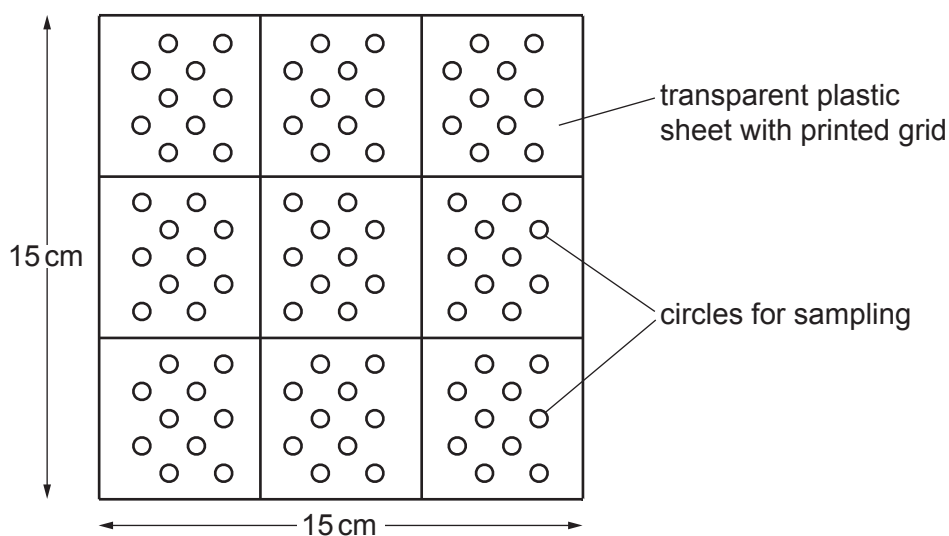


Fig. 2.2

For each tree sampled, the scientists:

- placed the grid against the tree trunk
- counted the number of circles that contained lichen
- used the number of circles that contained lichen to calculate the percentage cover of lichen on the tree
- measured the distance from the tree to the road.

(a) (i) State **two** variables that the scientists should standardise in this investigation.

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..... [2]

(ii) State **one** risk **and** the safety precaution that the scientists should take when measuring the abundance of the lichens on the tree trunks.

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- (b) The scientists recorded the presence of lichens by shading in the circles on the grid where lichens were observed.

The results are shown in Fig. 2.3.

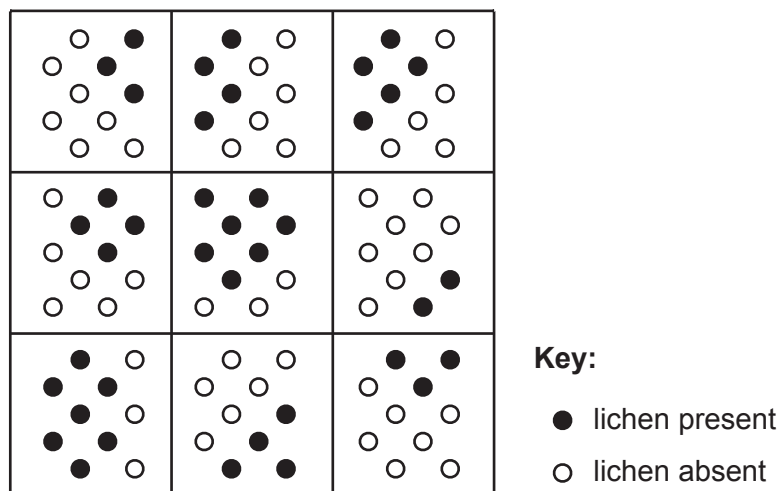


Fig. 2.3

Use the grid in Fig. 2.3 to calculate the percentage cover of the lichen on the tree trunk.

Show your working.

percentage cover of lichen = % [2]

- (c) The scientists measured the percentage cover of lichens on 10 trees, each growing at a different distance from the road.

The scientists carried out a Spearman's rank correlation analysis to find out if there was a correlation between the distance from the road and the percentage cover of lichens on the tree trunks.

Table 2.1 shows part of the table used to determine the data needed for the Spearman's rank correlation equation.

- (i) Complete Table 2.1 for tree number 10.

Table 2.1

tree number	distance from road /m	rank for distance from road	percentage cover of lichen	rank for percentage cover of lichen	difference in rank (D)	D^2
10	5.6	7	74	5		

[1]

The formula for calculating Spearman's rank correlation coefficient (r_s) is:

Key to symbols:

$$r_s = 1 - \left(\frac{6 \times \Sigma D^2}{n^3 - n} \right)$$

D = difference in rank between each pair of measurements

n = number of pairs of items in the sample

- (ii) The scientists calculated the value of ΣD^2 as **39.75**.

Calculate the Spearman's rank correlation coefficient (r_s) value to **three** significant figures.

Use this space for working.

Spearman's rank correlation coefficient (r_s) = [2]

- (iii) State the conclusions that can be made from the calculated value of Spearman's rank correlation coefficient (r_s).

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..... [3]

- (iv) State how the scientists could use the calculated value of r_s to determine whether or not the correlation is due to chance.

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..... [2]

[Total: 14]

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